
INTERFACE-INDUCED TRAJECTORY CENSORING: WHEN TOOL-CALL INTERFACES DISTORT EVALUA- TION AND RL EXPERIENCE

Anonymous authors

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ABSTRACT

Tool-use evaluations read their call rates from a serving stack. That rate can be zero while the model emits a complete call that the configured interface does not recognize. On Berkeley Function-Calling Leaderboard v4, holding weights, cases, decoding, seeds, executor, and scorer fixed, changing the serving adapter moves one model’s score under BFCL’s official scorer from 0.00 to 0.96 on `simple_python` and from 0.00 to 0.19 on `multi_turn_base`. A 2×2 over template and parser shows that neither one-component replacement recovers a call from the documented baseline, whereas replacing both recovers 196/200. The components are strongly complementary because they implement different serialization contracts. Across five Qwen2.5-Coder checkpoints under the mismatched interface, the server parses 0/100 calls at every size while independently classified call emissions rise from 0 to 80/100 (approximately 72 after human calibration). Under a matched envelope, a preregistered supporting ladder holds the silent fraction at 0–2 through 8B. On 115 stateful τ -bench retail tasks, the same adapter swap moves parsed and executed calls from 0 to 636 and tasks reaching any tool from 0 to 103. The mismatch also changes RL experience: in a preregistered 150-step 7B LoRA control whose arms differ only in the parser registered inside verl, the mismatched arm converts 8,689 tight-classified emissions into 10 executions, versus 16,844 executions after repair. Held-out rescues nevertheless remain 12 and final passes are 430/542 versus 427/542 ($p = 0.25$): the intervention restores tool-mediated experience, but we detect no held-out improvement at this scale, budget, and seed. The observed tool-call rate is not a property of the model alone; it is a property of the model–interface stack that measures it.

1 INTRODUCTION

Agent benchmarks and training loops rarely consume raw model generations. They consume an action produced by a chain: emission $\rightarrow$ serialization $\rightarrow$ parse $\rightarrow$ execution $\rightarrow$ observation $\rightarrow$ score or reward. A failure at several of these boundaries has the same observable as a model declining to act: HTTP 200 and an empty `tool_calls` array. We call the resulting measurement failure *interface-induced trajectory censoring*. The term covers two distinct effects: *masking*, where a tool-intended emission never enters the structured action channel, and *suppression*, where an interface constraint prevents an invalid action from entering that channel. Their common methodological problem is not a shared mechanism but a shared terminal observation that is easily attributed to the model. Figure 1 shows the layers measured in our experiments.

This problem first appeared as a puzzling RL run: reward rose, but almost all new passes occurred on turn 1 and multi-turn rescues stayed at 6–9 of 540. What we found was that the tool was never successfully called in those historical 1.5B rollouts. That observation alone was over-determined: a same-prompt probe found no correctly named call at that checkpoint, and the historical run did not retain full generations. We therefore separate what the model emits from what each downstream layer observes, and later construct a 7B single-variable training control that does retain the full evidence chain.

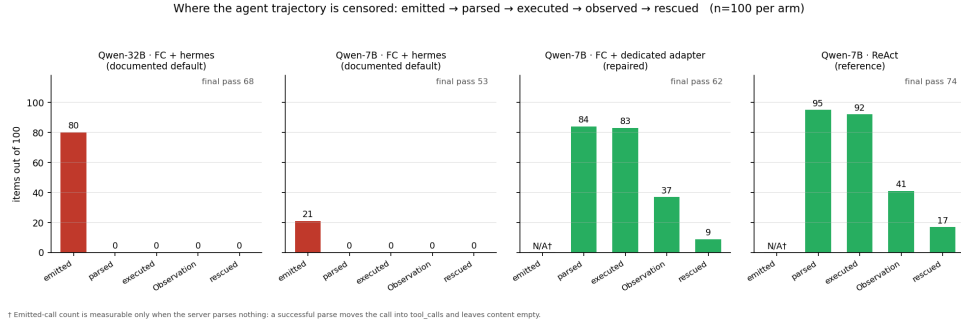


Figure 1: A tool-use measurement funnel. Calls present in raw generations can disappear before parsing, execution, observation, or scoring. N/A denotes cells that cannot be read from content once the server has moved a successful call into `tool_calls`.

The individual Qwen incompatibility was reported previously in vLLM issue #32926 (hanXen, 2026). Our contribution is to quantify it as an evaluation-validity failure, decompose its causes, and follow it into interactive evaluation and RL:

- On BFCL v4, a template–parser 2×2 shows zero recovery from either one-component replacement at the documented baseline and 196/200 parses when both are replaced; on τ -bench, the same joint repair restores 636 executions over 115 tasks.
- A five-checkpoint ladder measures a growing absolute undercount under one mismatched interface, while a preregistered matched-interface ladder keeps the silent fraction at 0–2.
- A paired Llama intervention shows the opposite interface effect: relative to the official-template baseline, `strict: true` suppresses 22 wrong-tool calls and raises final pass from 44 to 61.
- A preregistered 7B RL control changes only verl’s parser registry. Repair restores 16,844 executed and observed calls, but no held-out improvement is detected, separating access to experience from learning from it.

The resulting claim is narrower than “the stack matters.” A serving interface is part of both the benchmark’s measurement function and an agent’s effective action space. Consequently, tool-use results require layer-wise observability and a conformance check before model or protocol attribution.

2 RELATED WORK

Evaluation outcomes vary with prompt format, benchmark choice, and other nuisance factors (Sclar et al., 2024; Mizrahi et al., 2024; Dehghani et al., 2021). Tool-use audits similarly expose evaluator defects and ambiguous failure attribution (Bhat et al., 2026; Raj et al., 2026; Soni, 2026; Albayaydh et al., 2026). Those studies generally vary inputs or inspect traces above the serving boundary. We study a component inside the measurement apparatus: an independently configured serialization–parser contract that can erase an action without an error. The closest analogue is a security-log study where reparsing identical bytes changes measured accuracy from 0% to 76% (Garware & Zisad, 2026). We extend that structure to standard agent benchmarks, downstream execution, and RL experience. Harness-Bench argues for reporting model–harness configurations from cross-harness workflow measurements (Yao et al., 2026); we isolate one contract with component-level interventions and follow it into training. Agent-Reactive Bugs finds silent model–harness boundary failures in issue reports (Chen et al., 2026); our evidence is a controlled intervention rather than a bug-report taxonomy. BFCL (Patil et al., 2025), τ -bench (Yao et al., 2025), and other tool benchmarks (Qin et al., 2024; Li et al., 2023) supply the evaluation context; verl (Sheng et al., 2025), GRPO (Shao et al., 2024), and LoRA (Hu et al., 2022) supply the training stack. Constrained generation can hurt reasoning (Tam et al., 2024); our Llama control is a complementary case in which a constraint removes role confusion and improves execution.

3 FRAMEWORK AND TESTABLE SIGNATURES

Let Y be the assistant bytes produced from fixed weights, prompt, and decoding; $C(Y)$ an independent indicator that those bytes contain a tool-intended action; $P(Y)$ the structured action returned by the configured parser; and $E(P(Y))$ the execution and observation produced by the environment. A benchmark normally records a function of $P(Y)$ or $E(P(Y))$, not Y . We call an item *masked* when $C(Y) = 1$ but $P(Y) = \emptyset$. This definition avoids a metaphysical “true capability”: the directly observed fact is that a behavior exists in the raw generation and is absent from the structured trace. It also does not imply a parser bug. The parser may correctly reject bytes outside its grammar; the invalid measurement arises when checkpoint, template, and parser are treated as a compatible bundle without testing their contract.

The framework gives three diagnostic signatures. First, a parser-only intervention cannot change bytes already generated, but can change execution, observations, later turns, and rescues. Our evaluation-time Qwen repair changes parser, template, and few-shot together, so its identical aggregate turn-1 count (53 versus 53) is a consistency check, not layer identification. We instead localize the failure with fixed-output replay and event tracing. Llama’s `strict:true` intervention acts earlier by constraining generation and therefore can change turn-1 output.

Second, a size trend in $C(Y) - \mathbb{I}[P(Y) \neq \emptyset]$ requires both changing emissions and a fixed rejection boundary. Across a comparable scale span, our preregistered Qwen3 ladder under a matched envelope does not show the same growth. This supporting cross-family observation does not isolate family from envelope and is not a strict scale counterfactual. Third, in RL the parser changes the sampled experience distribution before a policy-gradient estimator sees it. Restoring actions and observations is therefore a causal claim about gradient *availability*, not about eventual learning. The formal 7B control tests the two claims separately: event counts identify experience access, while held-out checkpoints test whether that access changes behavior.

4 MEASUREMENT SETUP

Our controlled probe uses 100 KodCode problems (Xu et al., 2025), decontaminated against EvalPlus by normalized text/code n-gram containment, paired across arms, with one `run_tests(code)` tool. Function calling uses `tool_choice:auto`, temperature 0, and 2,048 output tokens. We separately run BFCL v4 at pinned Gorilla commit 6ea5797 and all 115 τ -bench retail tasks. The latter uses a locally served Llama-3.1-8B user simulator rather than the original harness’s `gpt-4o`; its absolute score is therefore not leaderboard-comparable, and we claim only paired arm differences.

Serving uses vLLM 0.27.1 (Kwon et al., 2023). Server-parsed calls are insufficient for the question under study. One shared classifier marks a generation *tight* only when it contains a JSON object naming `run_tests` and a literal Python program in `arguments.code`; schema echoes and placeholder identifiers are excluded. Against a third-party-adjudicated gold standard its κ is 0.871, and tight-stratum precision is 36/40=0.900. Two annotators achieve $\kappa = 0.967$ on the 62 items presented with identical bytes. Stored evaluation outputs have a 4,000-character cap, which creates false negatives, while 0.900 precision establishes false positives; the net bias is therefore not identified. In the stratified validation sample, cap hits by scale are 0/6, 0/13, 0/15, 0/24, and 10/40 from 1.5B through 32B, so truncation uncertainty is concentrated at the headline rung. Formal RL rollouts are retained in full with hashes. Full validation appears in Appendix F.

Before interpreting a zero we verify that `tool_choice:required` produces a correctly named, parseable call on the same live server. An arm enters pass-rate analysis only with 100 items, exit code 0, zero request errors, and matching provenance; this gate is enforced in code. Appendix A records an earlier invalid $p = 0.648$ that this rule forced us to withdraw.

For paired binary outcomes we use exact two-sided McNemar tests and report both discordant counts; for the formal RL endpoint we additionally report the paired risk difference with an approximate Wald interval using the variance of the within-item difference. With roughly twelve tests, the conservative Bonferroni threshold is $0.05/12 \approx 0.0042$. We distinguish a mechanism count (parse, execution, observation) from a task outcome throughout: a large deterministic movement in the former does not license a performance claim when the paired outcome test is null. BFCL scores come

from its official scorer; τ -bench outcomes come from final environment state; code-task outcomes execute the generated program against held-out tests. Training events exclude evaluation requests. Historical 1.5B curves exclude two verified EvalPlus defects (HumanEval/32 and Mbbp/599) and use $n = 540$. The preregistered 7B control retains all 542 items and scores those two as failures in both arms; they affect absolute rates but not paired discordances.

5 RESULTS

5.1 EXTERNAL VALIDITY: BFCL AND τ -BENCH

Holding model weights, cases, decoding, seeds, executor, and scorer fixed on BFCL v4, the documented Hermes configuration yields official scores 0.00 on both `simple_python` and `multi_turn_base`; the dedicated Qwen2.5-Coder adapter yields 0.96 and 0.19. Because the adapter changes both template and parser, we ran the full 2×2 (Table 1). Changing either component alone from the documented baseline recovers none (0/200); changing both recovers 196/200. Thus the two one-factor simple effects at that baseline are zero, while the joint replacement shows strong complementarity. This statement concerns parse counts; official BFCL scores were computed for the two end-to-end corner configurations, not all four cells. Each component behaves according to its own contract, but the contracts disagree. The documented arm has zero HTTP errors and 166/200 strict call emissions despite zero parses. On `multi_turn_base`, repair moves execution from 0 to 98/100 and success from 0 to 19/100. Complete scorer and funnel tables are in Appendix G.

Table 1: BFCL first-request parses, template $\times$ parser ($n = 200$).

	documented template	dedicated template
Hermes parser	0	0
Dedicated parser	0	196

On paired τ -bench tasks, the same adapter swap moves parsed and executed calls from 0 to 636 and tasks reaching any execution from 0/115 to 103/115. All 1,389 documented-arm requests carry tools; 789 assistant turns contain call-shaped payloads that are not parsed. Replaying vLLM’s extractor classifies all 789 as missing the required envelope, with zero malformed payloads and zero cases the parser should have accepted. Outcomes move from 7 to 10 solved, with three discordant pairs in the repaired direction ($p = 0.25$); we make no outcome claim. This result extends the envelope-layer mechanism to a non-code, stateful environment, but not the scale or RL claims, which remain code-only (Table 2).

Table 2: External benchmarks under documented and repaired adapters. BFCL rows are official scores; τ -bench rows are paired counts over 115 tasks.

measurement	documented	repaired
BFCL <code>simple_python</code> score	0.00	0.96
BFCL <code>multi_turn_base</code> score	0.00	0.19
BFCL multi-turn executed/succeeded	0/0	98/19
τ -bench parsed/executed calls	0/0	636/636
τ -bench tasks reaching a tool	0	103
τ -bench tasks solved	7	10

5.2 SCALE UNDER MISMATCHED AND MATCHED INTERFACES

Under the documented Hermes interface, the Qwen2.5-Coder server reports zero calls at all five sizes while tight-classified emissions rise monotonically from 0 to 80/100 (Table 3). The calibrated 32B estimate is approximately 72 if tight-stratum precision transfers across sizes. A random 300-item replication gives 0, 4, 30, 40, and 81 per 100: the 7B rung moves from 21 to 30, but direction

and monotonicity remain. This is a within-family observation under one mismatch, not a scaling law; the server column is constant, so the slope comes entirely from increasingly frequent or regular emissions. The 4,000-character cap creates false negatives, while classifier precision below one creates false positives; the raw count’s net bias is not identified.

The preregistered supporting ladder uses Qwen3 checkpoints up to 8B whose required envelope matches Hermes. The prediction “parsed > 0 at every size” was committed before inference. It holds, while the silent fraction stays 1, 1, 2, and 0 through 8B; the separate mismatched ladder reaches 80 at 32B. This cross-family observation is consistent with contract mismatch, rather than size alone, driving the undercount; it does not isolate the envelope because family and envelope change together. Parsed counts are not monotone, and we disabled thinking because the default exhausted the token budget before a call; both facts were recorded as protocol deviations. An attempted Granite rescue produced no calls under either Hermes or its native parser and is uninformative, not confirmatory.

Table 3: Scale ladders. “Silent” is a tight call emission not observed as a structured call.

family/interface	size	parsed	tight	silent
Qwen2.5-Coder/Hermes	1.5B	0	0	0
	3B	0	4	4
	7B	0	21	21
	14B	0	36	36
	32B	0	80	80
Qwen3/matched	0.6B	86	1	1
	1.7B	44	1	1
	4B	70	2	2
	8B	97	0	0

A within-lineage Qwen2.5-Instruct control further localizes the effect: under the same template, parser, items, and decoding, parsed rates are 1, 11, 63, 77, and 88 rather than Coder’s flat zero. After rerunning with a working executor, multi-turn rescues rise 0, 3, 6, 14, and 23 with that parsed column. The contrast is consistent with checkpoint-specific post-training differences, not parameter count or the shared family/template alone; it does not identify tool-format training as the unique cause. Across 4,254 de-duplicated first turns from all archived arms, replay finds zero cases that vLLM’s configured parser should have accepted but did not: this is contract mismatch, not a parser implementation bug (Table 4).

Table 4: Within-lineage Qwen2.5-Instruct control under the same interface and items. Rescues are final passes not passing on turn 1.

size	parsed	turn-1	final	rescues
1.5B	1	13	13	0
3B	11	14	17	3
7B	63	34	40	6
14B	77	38	52	14
32B	88	42	65	23

5.3 LLAMA: A CONSTRAINT SUPPRESSES THE WRONG ACTION

Llama-3.1-8B calls the function requested by the programming problem rather than `run_tests` on 23/100 items. A richer schema leaves 22 wrong calls (discordances 6/5, $p = 1.0$), a thought scaffold worsens them to 59 ($p = 2.84 \times 10^{-10}$), and the official template leaves 22 (1/0, $p = 1.0$). The previously quoted $p = 0.125$ belongs to the final-pass comparison, not wrong-tool labels. Adding `strict:true` to the official-template arm reduces wrong calls from 22 to 0 (22/0, $p = 4.77 \times 10^{-7}$), raises valid execution from 73 to 97, and final pass from 44 to 61, a 17-point single-variable change. The broader terse-FC to official-plus-strict comparison used in the ReAct decomposition is 49 to 61, hence 12 points for the interface bundle ($p = 0.0075$), followed by 19 for protocol

($p = 0.0019$). Thus the model exhibits role confusion under unconstrained function calling, while the interface determines whether that confusion reaches the environment. Appendix A corrects the corresponding initiation rate from 97 to 74. DeepSeek and Mistral only motivate additional template/token failure layers: the former never injects tools, and all four Mistral FC variants contain request errors, so neither supports a comparable performance claim. Table 5 reports the paired wrong-tool tests and their comparison baselines.

Table 5: Llama-3.1-8B paired controls. The first three interventions do not fix role confusion; `strict:true` does.

arm	wrong-tool calls	exact paired p (baseline)
terse FC	23	—
rich schema	22	1.000 (terse)
thought scaffold	59	2.84×10^{-10} (terse)
official template	22	1.000 (terse)
official template + strict	0	4.77×10^{-7} (official)

5.4 REPAIR CHANGES EVALUATION TRAJECTORIES

On Qwen2.5-Coder-7B, changing only the evaluation adapter moves parses 0→84, items reaching a second turn 0→37, rescues 0→9, and final pass 53→62. Aggregate turn-1 pass is 53 in both arms, while Llama’s schema intervention moves turn-1 pass from 34 to 46. Because the Qwen adapter also changes template and few-shot, this aggregate pattern is descriptive rather than a failure-layer locator; fixed-output replay and event tracing provide that localization. The Qwen final gain is not significant at $n = 100$ ($p = 0.093$) or after the random $n = 300$ replication and a Bonferroni threshold of approximately 0.0042 (160 → 178, nominal $p = 0.0385$). Repair restores the mechanism; we do not claim an evaluation outcome effect (Table 6).

Table 6: Qwen2.5-Coder-7B evaluation repair; only the serving adapter changes.

arm	parsed	≥ 2 turns	turn-1/final	rescues
FC + Hermes	0	0	53/53	0
FC + dedicated adapter	84	37	53/62	9
ReAct reference	95	40	57/74	17

5.5 PARSER REPAIR RESTORES RL EXPERIENCE; NO HELD-OUT IMPROVEMENT DETECTED

The RL experiment is a preregistered single-variable 150-step 7B GRPO comparison. Both arms use the same LoRA rank 32, data, mandatory prompt, seed, decoding, verifier, evaluation, and training budget; the sole intervention is verl’s registered parser. Full emissions, parser verdicts, executions, observations, and hashes are retained. Table 7 excludes held-out evaluation from training event counts and uses the same paired 542-item evaluation set at every checkpoint. Both step-0 evaluations use the same base checkpoint and evaluator; their equal aggregate counts are an observation, not evidence from which we infer the evaluation design.

Table 7: Formal 7B RL control. “Exec.” calls all return an observation.

arm	tight	accepted	exec.	final/rescues, step 0→150
broken FC	8,689	10	10	430→430 / 12→12
repaired FC	17,608	16,912	16,844	430→427 / 12→12

The 16,912 accepted calls occur in 16,844 generation events: 40 generations contain multiple calls and contribute 68 calls beyond the first. All 16,844 dispatches execute and return an observation; the complete cross-tabulation is in Appendix G.10.

The broken channel is nearly, not literally, closed: 10 of 8,689 tight-classified emissions become actions. Repair changes sampled tool experience by more than three orders of magnitude and returns an observation for every execution. It does not improve held-out outcomes: repaired FC ends at 427/542 versus 430/542 for broken FC (paired difference -0.55 percentage points; approximate paired 95% CI $[-1.18, 0.07]$; all three discordances favor broken; exact McNemar $p = 0.25$), and rescues remain 12. Relative to its own initialization, repaired FC has one gain and four losses ($p = 0.375$). We do not interpret these non-significant contrasts as harm. At this scale, budget, and single seed, parser repair restores tool-mediated experience, but we detect no improvement in held-out success or rescue counts. Held-out evaluation is identical across arms and deliberately bypasses both training parsers: the current LoRA is queried through verl’s native rollout RPC at temperature 0 for up to four 1,024-token programmatic-feedback turns, generated code is executed on the same 542 EvalPlus items, and test output is appended as a user message. This explains why the broken-training arm can still show 12 evaluation rescues. Each evaluation request attests the loaded global weight step.

The repaired job was interrupted after step 116 and resumed from a validated step-90 checkpoint; model, optimizer, scheduler, RNG, dataloader, source, configuration, and weight-step hashes were checked. The repeated step-90 evaluation differs by two final-pass items, reported as test–retest variation. Formal event totals use the final model’s lineage—original steps 1–90 plus resumed steps 91–150—and exclude the abandoned 26-update step-91–116 branch. Resume evidence and the available checkpoint endpoints are in Appendix G.10.

Two details prevent common but incorrect readings of this comparison. Both arms use LoRA because 7B full-parameter GRPO does not fit on one 80 GB GPU; this preserves the arm contrast but prevents strict comparison with the historical 1.5B full-parameter curves. Event totals also differ because repair changes trajectories: once calls execute, conversations contain more turns and hence more opportunities to emit another call. We therefore use rates within each arm only to describe its funnel, and use the single registered-parser intervention for causal attribution, not equality of raw denominators. The old ReAct arm with 23,676 executions and flat rescues is directionally consistent, but remains supporting evidence because it changes protocol as well as interface.

6 IMPLICATIONS AND LIMITATIONS

Measure the funnel. Reports should include emitted-call rate under a stated criterion, parser acceptance, dispatch/execution, observation return, request errors, adapter/template/parser versions and hashes, and task outcome. We provide a 98-line preflight that sends a canonical tools-bearing request, validates the returned name and arguments, and repeats under `tool_choice:required` as a positive control. It would catch each silent failure studied here. This should become a machine-readable conformance report rather than a model-only call rate.

At minimum, such a report should persist model and tokenizer revisions, template and parser hashes, decoding arguments, request errors, the emitted-intent criterion, acceptance, dispatch, execution and observation counts, and the outcome delta. Most fields already exist somewhere in a normal trace. The standard changes the unit of reporting from “model call rate” to a versioned model–interface measurement card, making silent zeros auditable and adapter changes comparable.

Interpretation limits. The scale and RL results use one code task and one tool; only the envelope mechanism is replicated on a non-code benchmark. The scale trend is one family under one mismatch, with single-sample headline arms and a 32B ceiling. The RL result is one 7B LoRA seed and budget, and near-zero samples do not establish zero policy probability. Roughly twelve McNemar tests imply a Bonferroni threshold near 0.0042; only the strongest interventions survive it, and we rest no claim on the nominal or null contrasts. A historical verifier used an unanchored pytest regex and omitted skipped/expected-failure outcomes; the formal arms use the same repaired verifier and retain full rollouts, so that flaw cannot explain their contrast. Results depend on vLLM 0.27.1, verl 0.9.0, and mutable publisher templates. Appendix H states all 17 limitations, including the non-random sample, post-treatment conditioning, asymmetric family taxonomy, human-validation scope, provenance corrections, and the observations that would change our conclusions.

Several of those limits bear directly on the headline. The original 100 items are `clean[:100]`, not a random draw; the random 300-item replication preserves the ordering but moves the 7B rung from

21 to 30 per 100. Conditioning protocol comparisons on items parsed by both arms is post-treatment selection, so the Qwen residual of +8.4 points (95% CI $[-0.5, 17.4]$, $p = 0.118$) is unresolved rather than evidence of equivalence. The taxonomy is intentionally asymmetric: Qwen and Llama have controlled repairs, while Mistral has no admissible 100-item FC run and DeepSeek only a template diagnosis. Complete RL attempt-versus-discard attribution exists only for the new 7B runs. The historical 1.5B data remain over-determined and are not used for the causal training claim.

What would change the conclusions. A replicated or longer repaired-FC run with a consistent rescue gain would narrow the scope of the present 150-step, single-seed null result. A second family whose undercount does not grow with size would confine the scale observation to Qwen-plus-Hermes; a comparable rise would strengthen the scaling interpretation. A non-code size ladder would separate the envelope mechanism already seen on τ -bench from the code-shaped payloads used in the current ladder. A second serving stack would identify which findings are vLLM-specific. These tests and decision rules are stated in full in Appendix H, rather than presented as evidence already obtained.

7 CONCLUSION

A tool-call rate measures a model–serialization–parser–execution composition. On BFCL, neither template nor parser is individually broken, yet their mismatch moves the official score from 0.00 to 0.96; in RL, changing only the parser moves observed executions from 10 to 16,844 without a detected held-out improvement. The practical consequence is simple: verify the contract and report each funnel layer before attributing a terminal zero, a protocol gap, or a training curve to the model.

AI USE STATEMENT

Generative AI systems assisted with experiment design feedback, implementation and debugging, result interpretation, literature discovery, and manuscript drafting and editing. The authors inspected the underlying trajectories and logs, reran validation checks, verified every reported number against persisted artifacts, and manually reviewed the final claims and citations. No AI-generated output is treated as experimental evidence. The authors take responsibility for all content, including code, claims, and text produced with AI assistance.

REPRODUCIBILITY STATEMENT

The anonymous supplementary material contains executable analysis code, pinned component versions, configuration and source hashes, arm-level provenance, preregistrations committed before inference, full statistical outputs, and the retained rollout events used for the RL audit. Appendix B provides a minimal reproduction path; Appendices D–G provide configurations, arm inventories, validation records, and complete result tables. Appendix A records every known provenance or instrumentation error and its effect on the reported conclusions.

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A DATA ERRATA

Every number in this paper traces to a trajectory file under `runs/final/`. This appendix lists the files whose recorded provenance is wrong, the arms that are inadmissible for pass-rate comparison, and what was verified correct. It is reproduced from `runs/final/ERRATA.md` in the released repository.

A.1 SEVEN REACT ARMS RECORD THE WRONG MAX_TOKENS

The following files record `max_tokens=2048` while the generation limit in force was **1024**. Cause: `gen()` in the probe carried a default of `max_tokens=1024` while `gen_fc()` used the global `MAX_TOKENS=2048`, and the provenance written to disk recorded `MAX_TOKENS` for both. Fixed 2026-08-31.

<code>traj_v3_Llama8B_react.jsonl</code>	recorded 2048 / actual 1024
<code>traj_v3_Qwen7B_react.jsonl</code>	recorded 2048 / actual 1024
<code>traj_v3_Mistral7B_react.jsonl</code>	recorded 2048 / actual 1024
<code>traj_v11_DS1.3b_react.jsonl</code>	recorded 2048 / actual 1024
<code>traj_v11_DS6.7b_react.jsonl</code>	recorded 2048 / actual 1024
<code>traj_v11_Llama32_1B_react.jsonl</code>	recorded 2048 / actual 1024
<code>traj_v11_Llama32_3B_react.jsonl</code>	recorded 2048 / actual 1024

Direction of the bias. The contemporaneous FC arms were genuinely at 2048, so in every affected ReAct-vs-FC comparison **the function-calling side had the larger generation budget**. ReAct nonetheless won those comparisons, so the reported protocol gaps are *conservative*; correcting the asymmetry can only widen them. Re-running the three main ReAct arms at a true 2048 changed final pass by at most one item ($80 \rightarrow 80$, $74 \rightarrow 74$, $32 \rightarrow 33$), which is why the correction is small in practice.

Replacement data. The four v11 arms are superseded by `traj_v13*_react.jsonl`; the three v3 arms by `traj_v14*_react.jsonl`. All tables in this paper cite the v13/v14 versions; the v3/v11 ReAct arms are retained only as a record.

A.2 LLAMA-3.1-8B INITIATION RATE REQUIRED MANUAL CORRECTION

`traj_v3_Llama8B_fc.jsonl` records an initiation rate of 97/100, but 23 of those calls target **the task function rather than `run_tests`**. The probe did not verify tool names when that batch was collected, so `has_action` did not distinguish them. **The corrected `run_tests` initiation rate is 74/100.**

A re-run restricted to exactly those 23 items under matched configuration (`traj_v8_Llama8B_recheck.jsonl`, seed fixed) reproduced 23/23 as wrong-tool calls, with names such as `can_form_word`, `check_password_strength` and `floyd.warshall`.

Of the same batch, `traj_v3_Qwen7B_fc_nosuffix.jsonl` has an initiation rate of 0 and needs no correction; `traj_v3_Mistral7B_fc.jsonl` has 2, and those two were not individually re-checked for tool name.

A.3 ARMS INADMISSIBLE FOR PASS-RATE COMPARISON

traj_v3_Mistral7B_fc.jsonl	n_err = 2	(rc=2)
traj_v5_Mistral7B_fc_official.jsonl	n_err = 42	(rc=2)
traj_v9_Mistral7B_fc_strict.jsonl	n_err = 3	(rc=2)
traj_v9_Mistral7B_fc_official_strict.jsonl	n_err = 39	(rc=2)
traj_v13_Llama8B_fcstrict_t06_s3.jsonl	n_err = 1	(rc=2)

For these arms the **error-rate census remains valid** — that census is itself a result — but pass rates are not comparable because data are missing, and no p -value is computed for them. An earlier draft computed $p = 0.648$ on one such arm; that number is retracted.

A.4 PROVENANCE SCHEMA DRIFT

Fields were added over the course of the work, giving four schema generations: early arms lack `seed`, `temperature` or `fc_schema`. The fields load-bearing for every claim (`model`, `protocol`, `adapter`, `max_tokens`, `clean_index`) are present in all generations. A `script_sha256` field was never successfully added — the patch was interrupted twice — so script identity across a run is instead guaranteed externally, by hashing the probe before launch (`v13_pinned_hashes.txt`) and re-checking after.

A.5 VERIFIED CORRECT

- All 50 full-length arms use the identical 100-item set (`clean[:100]`; unique item-set count = 1), so every cross-arm comparison in this paper is exactly paired.
- Every FC arm ran at a true `max_tokens` of 2048.
- Sampling in the variance experiment is genuine: at `temperature = 0.6`, two seeds differ in first-turn code on 95/100 items and flip outcomes on 16/100.

A.6 THIRD-PARTY COMPONENT VERSIONS

The Qwen2.5-Coder `<tools>` parser is taken from `hanXen/vllm-qwen2.5-coder-tool-parser` (Apache 2.0), downloaded 2026-08-31, at the then-current main:

```
commit 1b921501f30cbfe347dccbl7de3c82a1d55131
(1b92150, 2026-04-29)
"fix: buffer partial <tools> prefix in streaming
to prevent tag leak"

SHA256 (first 20):
c16bb1f88936a2d96c7c  qwen2_5_coder_tool_parser.py
                        (unmodified)
736bd175adbf90942c1d  tool_chat_template_qwen2_5_coder.jinja
                        (MODIFIED)
a95b2a9b91e65b3d452f  tool_chat_template_qwen2_5_coder.jinja.orig
                        (original)
```

The template was modified because its few-shot examples invoke `get_weather`, a tool absent from our schema. Left unchanged it would induce calls to a non-existent tool, simultaneously inflating the initiation count and contaminating the empty-argument statistic. The original is retained alongside.

B REPRODUCIBILITY

```
# analysis and figures (no GPU)
pip install -r requirements.txt
python analysis/intent.py           # the single intent criterion, all four counts
python runs/final/finaltable.py    # main / variance / error-rate tables (N/A enforced)
python figures/make.figs.py

# probe (needs a vLLM server)
```

```
python probe_react_full.py --model <path> --port 8000 --n 100 \
  --protocol {react,fc} --strength {optional,mandatory} \
  --fc-schema {terse,rich,strict} --parser-adapter cross.family \
  --temperature 0.0 --seed 0 --out traj.jsonl
python validate_arms.py # per-arm admissibility: lines / rc / n_err / provenance /
```

Model snapshots must be pinned: `tokenizer_config.json` inheritance is the mechanism behind §5.3’s false-positive capability check, so a family’s behaviour can change with a repository revision. Every model path, its HF revision, and the serving flags used are recorded per arm in the trajectory provenance and in `runs/final/by-config/README.md`.

Trajectories for all 50 full-length arms, the training logs, the errata, and the per-configuration index are in `runs/final/`. Third-party parser: `hanXen` (1b92150, Apache 2.0), few-shot examples rewritten from `get_weather` to `run_tests`; original retained with both hashes recorded.

C PROMPTS AND TOOL SCHEMAS: ENGLISH RENDERINGS

The experiments used Chinese prompts. To keep the submission manuscript entirely in English, this appendix gives faithful English renderings; the byte-exact originals and their hashes are included in the anonymous supplementary artifact. The renderings below were never sent to a model and are provided only for review.

C.1 REACT PROMPTS

```
REACT_OPTIONAL
-----
You are a Python coding assistant. You may work in the following format:

Thought: your reasoning
Action: run_tests
Action Input: ```python
<complete code>
```

Observation: (filled in by the system)

Thought: I have confirmed that the code is correct
Final Answer: ```python
<final code>
```

You may submit code for verification with Action, or answer directly.
```

The mandatory variant replaces the final sentence with: “You must first submit code for verification with Action, then give a Final Answer,” and marks the loop as repeatable. The main evaluation uses the optional prompt for both protocols. Historical FC-ReAct training and both formal P3 arms use mandatory prompts of matched strength.

C.2 FUNCTION-CALLING PROMPTS

```
FC_OPTIONAL
-----
You are a Python coding assistant. Implement the function described by
↪ the user.
You may call run_tests to submit your code to the test runner. It returns
pass/fail status and errors, which you may use to revise and resubmit.

FC_MANDATORY
-----
You are a Python coding assistant. Implement the function described by
↪ the user.
You must first call run_tests to verify your code; do not answer
↪ directly.
```

The tool returns pass/fail status and errors. If any test fails, revise
↪ the
code and call `run_tests` again.

C.3 TERSE, RICH, AND STRICT SCHEMAS

The terse schema is the documentation-level baseline. Its English rendering is:

```
{ "type": "function", "function": {  
  "name": "run_tests",  
  "description": "Submit the Python code you wrote to the test runner; it  
    returns pass/fail status and errors.",  
  "parameters": { "type": "object",  
    "properties": { "code": { "type": "string" } },  
    "required": [ "code" ] } } }
```

The rich control adds: “This tool is not the function you are being asked to implement. Do not pass the task function’s parameters into it. Its only parameter, `code`, is the complete source text you wrote,” plus the example `def add(a,b): return a+b`. Under this warning, 22/100 items still pass the task function’s parameters (6/5 discordances, exact $p = 1.000$ versus terse). The strict schema is terse plus `"strict": true` and `additionalProperties:false`, with strict tool calling enabled. This single change takes the wrong-tool count to zero.

C.4 THOUGHT SCAFFOLD AND ROLE-DISAMBIGUATION CONTROL

The Thought-scaffold prefix reads: “First write your reasoning in a Thought: what the problem asks, its edge cases, and your intended algorithm. Once you have thought it through, call the `run_tests` tool.” Nothing else changes; wrong-tool calls rise from 23 to 59 ($p < 0.001$).

The 1.5B role-disambiguation control adds the following contrast to FC.MANDATORY:

Common mistake: the function requested by the user is not a callable
↪ tool.
The only tool is `run_tests`, whose sole argument `code` contains your full
↪ source.

Wrong (task asks for `join_integers`):
{ "name": "join_integers", "arguments": { "int_array": [1,2,3] } }

Right:
{ "name": "run_tests", "arguments": { "code": "def join_integers(int_array):\n return ','.join(map(str, int_array))" } }

Recoverable call-shaped outputs rise from 52 to 64, but correctly named calls remain zero and final pass falls from 15 to 13.

C.5 INTENT CRITERION

Every emitted-call count uses one implementation without per-arm tuning:

```
TIGHT = re.compile(  
  r' "name"\\s*:\\s*"run_tests".{0,200}? "arguments"\\s*:\\s*\\{ '  
  r'.{0,80}? "code"\\s*:\\s*"(.{0,4000}?)"\\s*\\}', re.S)  
REAL = re.compile(r'\\n|def |class |return |import |lambda |'  
  
def is_tight_call(raw):  
  m = TIGHT.search(raw or "")  
  return bool(m and REAL.search(m.group(1)))
```

The criterion requires the tool name, an `arguments` object, and a string-valued `code` field containing Python-like material. It excludes schema echoes, identifiers passed instead of strings, and placeholders. The strong tier drops the payload requirement; the weak tier counts JSON-shaped fragments without the tool name and is disjoint from both. Validation is in Appendix F.

D SERVING AND TRAINING CONFIGURATION

D.1 PER-FAMILY SERVING FLAGS

These are the configurations under test, including documented serving flags and the adapter/schema interventions identified in our experiments. The rightmost column summarizes observed limitations or failure symptoms; it is not a uniform flag-omission ablation.

Table 8: Serving configurations and observed symptoms (vLLM 0.27.1).

Family	Flags / intervention	Observed symptom
all	<code>--enable-auto-tool-choice</code>	HTTP 400 if omitted
Qwen2.5 (Instruct)	<code>--tool-call-parser hermes</code>	silent empty <code>tool_calls</code>
Qwen2.5-Coder	<i>no working stock configuration; we install the community <tools> parser plus its chat template</i>	silent, at every scale
Llama 3.1 / 3.2	<code>--tool-call-parser llama3.json</code> <code>--chat-template tool_chat_template_</code> <code>llama3.1.json.jinja</code> plus "strict": true in the schema	malformed arguments; without strict the terse baseline has 23% wrong-tool calls
Mistral (HF format)	<code>--tokenizer-mode hf --config-format hf --load-format hf</code> <code>--tool-call-parser mistral</code> <code>--chat-template tool_chat_template_</code> <code>mistral_parallel.jinja</code> <code>tool_call_id</code> must be exactly 9 characters	HTTP 400; the officially recommended parallel template <i>raises</i> the error rate from 2% to 42%
DeepSeek-Coder	none available: the chat template never injects tools	n/a

Missing auto-tool-choice support and the tested Mistral configurations produce explicit request errors. Other mismatches can return HTTP 200 with empty `tool_calls`, indistinguishable at that field from declining to call a tool. The released 98-line `preflight_toolcall.py` checks a canonical request for a non-empty call with the right name and parseable arguments, then repeats under `tool_choice: required` as a positive control. This is a screening check, not a guarantee of compatibility on all tasks.

D.2 TRAINING CONFIGURATION

The historical 1.5B runs use the following configuration; they motivate the investigation but are not the formal parser-repair control:

Table 9: Training configuration. All arms share every value below; arms differ only in the protocol and, for the historical runs, the algorithm and seed.

Setting	Value
Model	Qwen2.5-Coder-1.5B-Instruct
Framework	verl 0.9.0 (vLLM 0.27.1 rollout backend)
Algorithm	GRPO ($\times 2$ seeds) and RLOO ($\times 1$ seed)
Trajectories per step	<code>train_batch_size</code> $16 \times \text{rollout.n}$ $8 = 128$
Steps	150 for the historical runs and formal 7B control; 10 / 3 / 75 for the historical diagnostic arms in Appendix G
Training data	KodCode, decontaminated pool of 7669 items
Held-out evaluation	EvalPlus: HumanEval+ 319, MBPP+ 677
Denominators	540 (multi channel), 454 (repair channel), after removing 2 known-defective items
Agent loop	<code>tool_agent</code> (FC) or <code>react_agent</code> (ReAct)
Tool	<code>run_tests</code> , single code argument, sandboxed <code>pytest</code>
System prompt	FC_MANDATORY / ReAct mandatory — matched strength (Appendix C)
Hardware	single A800 80 GB

The formal control of §5.5 instead uses Qwen2.5-Coder-7B-Instruct, GRPO, LoRA rank and alpha 32 over all linear modules, seed 0, 150 steps, and the same 16 prompts $\times$ 8 rollouts per step in both arms. Both use the mandatory FC prompt and fixed verifier, retain complete rollout text, and evaluate the same 542 paired multi-turn items at steps 0/30/60/90/120/150. Evaluation uses a shared parser-independent programmatic-feedback protocol: the current LoRA is queried through verl’s native rollout RPC at temperature 0, for at most four turns and 1,024 tokens per turn; generated code is executed against EvalPlus tests and the observation is appended as a user message. Native RPC metadata must match the expected global weight step. The sole arm difference is `multi_turn.format=hermes` versus the registered `qwen2_5_coder` parser. Full-parameter 7B RL did not fit the single 80 GB card, so both arms use the same parameter-efficient configuration; this preserves the arm contrast but prevents a strict comparison to the historical full-parameter curves.

D.3 TOOL-CALL INSTRUMENTATION

verl 0.9.0’s step metrics contain **no** tool-call count. The metric named `timing_s/agent_loop/tool_calls` is *elapsed seconds*; reading it as a count is a hard error and we flag it because the name invites exactly that.

We therefore instrumented the two agent-side call sites directly, each writing one labelled line per call:

- `CodeTool.execute` — the function-calling path
- `ReActAgentLoop._run_tests` — the ReAct path

The counter must not be placed in `sandbox.run_tests`. The reward function evaluates the final program through that same function, so a counter there conflates reward evaluation with agent tool use. We hit this and moved the counter; the sandbox’s own counter was renamed `sandbox.exec_count` so the two can never be confused again.

In the historical diagnostic run, the FC counter file has **0 lines**; the ReAct arm’s has **788**, all carrying the `react` label, so the provenance is uncontaminated. Three mutually independent signals agree — `num_turns` pinned at its minimum of 2, tool time exactly 0.00 s, and a dedicated counter at 0.

D.4 CONSTRUCTING THE REPAIRED-FC TRAINING ARM

The control initially appeared blocked at three layers:

1. `multi_turn.format=hermes` selects **verl’s own** `ToolParser` registry, which is independent of vLLM’s — so installing a vLLM parser plugin has no effect on the training path.
2. A verl-side parser we wrote does recover calls, but **none of them are the tool**: 36/52 target the task function, 16/52 carry executable code elsewhere in the body. Name normalisation cannot repair this, because no call carries code to normalise.
3. The 1.5B checkpoint emits no correctly named call for that parser to accept, and 7B full-parameter training does not fit a single 80 GB card.

The first two observations determine the intervention; the third determines the scale and LoRA configuration. We registered the parser inside verl and ran the matched 7B pair. Thus the earlier statement that no executable repaired-FC arm existed was a description of the configuration surface, not a permanent limitation; §5.5 reports the completed control.

D.5 THE PATH TO THE REPAIRED-FC TRAINING ARM

We encountered four successive obstacles while constructing the control. Each attempt is a measurement, so we retain the record; all were eventually resolved by registering the parser in verl and moving to matched 7B LoRA arms.

(i) A vLLM parser plugin does not reach the training loop. verl’s `AgentLoop` performs its own tool-call parsing (`verl/experimental/agent_loop/tool_parser.py`, registered as

hermes, adapted from vLLM v0.9.1). `multi_turn.format=hermes` selects *that* parser, not the server’s, so attaching a plugin to the vLLM rollout server changes nothing in training — a distinction worth stating because the two look identical from the outside.

(ii) **A verl-side parser recovers calls, but none of them are the tool.** We implemented a `qwen2.5_coder` parser inside verl’s registry accepting `<tools>` tags, json fenced blocks and bare JSON. On the exact training condition (1.5B, mandatory prompt, $n=100$) it lifts recoverable calls from **0/100 (hermes) to 52/100**. Of those 52, **zero name** runtests (Table 31). Every call targets the function the task asks the model to *write* — `can_form_word({"tiles", "word"})`, `check_password_strength({"password"})` — the same signature as Llama’s failure in §5.3, now at a $5\times$ smaller model. **Name normalisation cannot repair this: no call carries code to normalise.**

(iii) **A role-disambiguation few-shot makes it worse, not better.** Since the failure is semantic rather than syntactic, we targeted it directly: a system prompt containing the wrong call and the right call side by side, with an explicit warning that the task function is not a tool. Same model, same 100 items, same server; only the system prompt changes. The intervention made the model *more fluent at emitting the wrong call* ($52 \rightarrow 64$ recoverable) and slightly worse at the task ($15 \rightarrow 13$), with the count that matters staying at zero (Table 32). This is the third independent instance of the pattern in §G.9.

(iv) **Constrained decoding is unavailable in this stack.** Forcing the tool choice, or schema-constrained decoding, is not exposed by verl 0.9.0’s vLLM rollout path (its only `strict` key belongs to the profiler). This remains true but was the wrong diagnosis of what the repaired arm needed: guided decoding forces a format, whereas the successful intervention accepts the model’s existing format. See §5.5.

E ARM INVENTORY

Every full-length arm, generated from the trajectory files’ own provenance rather than transcribed by hand. All 50 arms share one item set (`clean[:100]`; distinct-set count verified = 1), so every cross-arm comparison in the paper is exactly paired. Fields recorded as --- were absent from that generation’s provenance schema (Appendix A.4); this is disclosed rather than back-filled. Seven arms carry a known-wrong `max_tokens` record and are listed in Appendix A.1; five are inadmissible for pass rates and are listed in Appendix A.3.

Not listed: `v8.Llama8B_recheck` (23 items, a targeted qualitative re-run) and eight `*smoke*` files at $n=3$. Neither is a formal arm.

Table 10: All 50 full-length arms and their recorded configuration.

Arm	Model	Prot.	Strength	Schema	Adapter	tok	temp	seed
a14.Qwen1.5B_fc_mandatory	Qwen-1.5B	fc	optional	terse	cross_family	2048	—	—
roleprobe.Qwen1.5B_fc_roledisambig	Qwen-1.5B	fc	optional	terse	cross_family	2048	0.0	0
v11_DS1.3b_react	DS-1.3B	react	optional	terse	cross_family	2048	—	—
v11_DS6.7b_react	DS-6.7B	react	optional	terse	cross_family	2048	0.0	—
v11.Llama32.1B_fc_strict	Llama-3.2-1B	fc	optional	strict	cross_family	2048	0.0	—
v11.Llama32.1B_react	Llama-3.2-1B	react	optional	terse	cross_family	2048	0.0	—
v11.Llama32.3B_fc_strict	Llama-3.2-3B	fc	optional	strict	cross_family	2048	0.0	—
v11.Llama32.3B_react	Llama-3.2-3B	react	optional	terse	cross_family	2048	0.0	—
v13_DS1.3b_react	DS-1.3B	react	optional	terse	cross_family	2048	0.0	—
v13_DS6.7b_react	DS-6.7B	react	optional	terse	cross_family	2048	0.0	—
v13.Llama32.1B_react	Llama-3.2-1B	react	optional	terse	cross_family	2048	0.0	—
v13.Llama32.3B_react	Llama-3.2-3B	react	optional	terse	cross_family	2048	0.0	—
v13.Llama8B_fcstrict.t06.s1	Llama-3.1-8B-Instruct	fc	optional	strict	cross_family	2048	0.6	1
v13.Llama8B_fcstrict.t06.s2	Llama-3.1-8B-Instruct	fc	optional	strict	cross_family	2048	0.6	2
v13.Llama8B_fcstrict.t06.s3	Llama-3.1-8B-Instruct	fc	optional	strict	cross_family	2048	0.6	3
v13.Llama8B_react.t06.s1	Llama-3.1-8B-Instruct	react	optional	terse	cross_family	2048	0.6	1
v13.Llama8B_react.t06.s2	Llama-3.1-8B-Instruct	react	optional	terse	cross_family	2048	0.6	2
v13.Llama8B_react.t06.s3	Llama-3.1-8B-Instruct	react	optional	terse	cross_family	2048	0.6	3
v14.Llama8B_react	Llama-3.1-8B-Instruct	react	optional	terse	cross_family	2048	0.0	—
v14.Mistral7B_react	Mistral-7B	react	optional	terse	cross_family	2048	0.0	—
v14.Qwen7B_react	Qwen-7B	react	optional	terse	cross_family	2048	0.0	—
v3.Llama8B_fc	Llama-3.1-8B-Instruct	fc	optional	—	cross_family	2048	—	—
v3.Llama8B_fc_legacy	Llama-3.1-8B-Instruct	fc	optional	—	cross_family	2048	—	—
v3.Llama8B_react	Llama-3.1-8B-Instruct	react	optional	—	cross_family	2048	—	—

Arm	Model	Prot.	Strength	Schema	Adapter	tok	temp	seed
v3_Mistral7B_fc	Mistral-7B	fc	optional	terse	cross_family	2048	—	—
v3_Mistral7B_react	Mistral-7B	react	optional	terse	cross_family	2048	—	—
v3_Qwen7B_fc_legacy	Qwen-7B	fc	optional	terse	cross_family	2048	—	—
v3_Qwen7B_fc_nosuffix	Qwen-7B	fc	optional	terse	cross_family	2048	—	—
v3_Qwen7B_react	Qwen-7B	react	optional	terse	cross_family	2048	—	—
v4_Llama8B_fc_cot	Llama-3.1-8B-Instruct	fc	optional	terse	cross_family	2048	—	—
v4_Llama8B_fc_rich	Llama-3.1-8B-Instruct	fc	optional	rich	cross_family	2048	—	—
v5_Llama8B_fc_official	Llama-3.1-8B-Instruct	fc	optional	terse	cross_family	2048	—	—
v5_Mistral7B_fc_official	Mistral-7B	fc	optional	terse	cross_family	2048	—	—
v5_Qwen1.5B_fc_intent	Qwen-1.5B	fc	optional	terse	cross_family	2048	—	—
v5_Qwen14B_fc_intent	Qwen-14B	fc	optional	terse	cross_family	2048	—	—
v5_Qwen32B_fc_intent	Qwen-32B	fc	optional	terse	cross_family	2048	—	—
v5_Qwen3B_fc_intent	Qwen-3B	fc	optional	terse	cross_family	2048	—	—
v5_Qwen7B_fc_intent	Qwen-7B	fc	optional	terse	cross_family	2048	—	—
v6_Llama8B_fc_strict	Llama-3.1-8B-Instruct	fc	optional	strict	cross_family	2048	—	—
v6b_Qwen1.5B_fc_plugin	Qwen-1.5B	fc	optional	terse	cross_family	2048	—	—
v6b_Qwen3B_fc_plugin	Qwen-3B	fc	optional	terse	cross_family	2048	—	—
v6b_Qwen7B_fc_plugin	Qwen-7B	fc	optional	terse	cross_family	2048	—	—
v9_Mistral7B_fc_official_strict	Mistral-7B	fc	optional	strict	cross_family	2048	—	—
v9_Mistral7B_fc_strict	Mistral-7B	fc	optional	strict	cross_family	2048	—	—
x2_DS1.3B_react	DS-1.3B	react	optional	—	—	—	—	—
x2_DS6.7B_react	DS-6.7B	react	optional	—	—	—	—	—
x2_Llama8B_fc	Llama-3.1-8B-Instruct	fc	optional	—	—	—	—	—
x2_Llama8B_react	Llama-3.1-8B-Instruct	react	optional	—	—	—	—	—
x2_Mistral7B_fc	Mistral-7B	fc	optional	—	—	—	—	—
x2_Mistral7B_react	Mistral-7B	react	optional	—	—	—	—	—

F HUMAN VALIDATION OF THE INTENT CRITERION, IN FULL

The scale trend in §5.2 rests on an automated classifier, so we validated it against human judgement. 98 outputs were sampled **stratified by classifier verdict** (40 *tight*, 28 *strong-but-not-tight*, 30 *neither*) — deliberately over-sampling the decision boundary so both false positives and false negatives are estimable. Sampling is seeded and the annotator saw only the raw output, never the classifier’s label. We report **two rounds**, because the first is itself a finding.

Table 11: Human validation of the intent criterion, round 1 and its adjudication. The classifier is used only to count what the server discarded, so its recall matters more than its precision.

	agreement	Cohen’s κ	precision	recall	FP	FN
Round 1 (raw output only)	86.7%	0.713	70.0%	96.6%	12	1
After adjudication	96.9%	0.936	95.0%	97.4%	2	1

The 12 round-1 false positives split into two causes, and the second is ours, not the annotator’s. **Eleven are reading-order failures:** the tool call sits mid-document, after a fenced `python` block (match positions 1209–2585 in outputs of median length ~ 2000); the annotator read the code block, concluded “this is a direct answer, not a call”, and stopped. **One is an instrument failure:** for a 32B item the matched span begins at character 2585 and the annotation pack truncated every output at 2400, so the evidence was *not in the pack at all* and that item could not have been labelled correctly. Re-shown the matched span alone, 10 of 13 disputed items were reversed. The corrected pack carries every output in full.

The first adjudication was one-sided, and we do not treat its κ as a reliability estimate. Auditing the procedure after the fact: the 13 adjudicated items were selected *exactly* as the items where the annotator disagreed with the classifier (13/13 overlap), and all 10 reversals moved *toward* the classifier, none away. Under that selection rule agreement can only rise, so $0.713 \rightarrow 0.936$ is a property of which items were re-examined, not evidence of reliability. We report the second annotator instead.

A SECOND INDEPENDENT ANNOTATOR

A second annotator, blind to the classifier, to the first annotator’s labels and to the first pack’s item order, labelled the identical 98 items under a verbatim-identical rubric. Two changes were made to the instrument: outputs are carried **in full**, and the adjudication rule was fixed in advance to review *all* three-way disagreements rather than only those disagreeing with the classifier.

Table 12: Inter-annotator reliability. Of 98 sampled items, 97 have complete paired A1/A2 records. The identical-byte subset removes a presentation confound introduced by the first pack’s 2,400-character cap.

Cohen’s κ (95% CI, bootstrap)	all items ($n=97$)	identical bytes ($n=62$)
A1 vs. A2 (inter-annotator)	0.736 [0.590, 0.863]	0.967 [0.897, 1.000]
A2 vs. classifier	0.936 [0.855, 1.000]	0.967 [0.897, 1.000]
A1 vs. classifier	0.712 [0.559, 0.847]	0.934 [0.834, 1.000]

Of the 12 paired disagreements, 11 have full text exceeding 2,400 characters and A1 saw a different presentation from A2; the first-round audit attributed most errors to reading order and one to evidence actually omitted by truncation. We therefore do not assign all 11 to one cause. On the 62 items where both read identical bytes the two agree on 61, and **inter-annotator $\kappa = 0.967$** .

Third-party adjudication, and the final labels. Fifteen items were not unanimous across A1, A2 and the classifier. All fifteen went to a third adjudicator under the pre-registered rule — selected by *any* three-way disagreement, never by disagreement with the classifier — who saw full text and no one else’s labels. The adjudicator sided with A2 on 11, with the classifier on 9, and with A1 on 6, and **moved away from the classifier on 6 of 15**. That last number is the check that matters: the first round’s adjudication moved away from the classifier zero times out of thirteen, which is what an outcome looks like when the selection rule has already decided it. Combining the 83 unanimous items with the 15 adjudicated ones gives a 98-item gold standard. The third adjudicator was blind to classifier and annotator labels; the classifier nevertheless affected the original stratified sample and which non-unanimous items required adjudication, so we do not call the gold set classifier-independent.

Table 13: Validation against the adjudicated gold standard. Pairwise denominators differ because one sampled item lacks a complete A1/A2 pair; classifier–gold uses all 98 items.

Cohen’s κ (95% CI, bootstrap)	full n	full sample	identical bytes ($n=62$)
A1 vs. A2 (inter-annotator)	97	0.736 [0.595, 0.866]	0.967 [0.893, 1.000]
Classifier vs. gold standard	98	0.871 [0.756, 0.958]	0.934 [0.829, 1.000]

Correction factor. Against the gold standard the tight stratum’s precision is $36/40 = 0.900$, so §5.2’s headline 80/100 at 32B corrects to ≈ 72 . The criterion errs in both directions — 4 over-counts against 2 calls it misses entirely, a net over-count of 2 — so the correction is smaller than the precision figure alone suggests. **The trend is unaffected:** applying 0.900 uniformly gives 0, 3.6, 18.9, 32.4, 72 against a server-side zero at every size. We report uncorrected classifier counts in all tables with this factor stated, rather than rescaling silently.

For the record, this number moved twice as the validation improved, and we report the trajectory rather than only its endpoint: **0.950** under the first, one-sided adjudication; **0.975** under A2’s labels alone; **0.900** against the adjudicated gold standard. The first was inflated by its selection rule, the second rested on a single rater, and only the third is built from a rule fixed before the labels were seen.

A bound on the measurement itself. The probe stores at most 4000 characters of raw output per item; 10 of the 98 sampled outputs reach that ceiling, and one grey-zone item is a call truncated by it. The classifier and both annotators read identical bytes, which is what agreement requires, but a call emitted beyond character 4000 is invisible to all three. All ten cap hits occur in the 32B subset of this stratified sample: by increasing scale the hit counts are 0/6, 0/13, 0/15, 0/24, and 10/40. These are sample diagnostics, not population truncation rates, but they show that the uncertainty is concentrated at the headline rung. The cap therefore creates downward pressure through false negatives. Because the classifier also has false positives (tight precision 0.900), however, the net bias of the raw count is not known; we report both the uncorrected count and the adjudication-based calibration rather than calling the total a strict lower bound.

G SUPPLEMENTARY RESULTS

G.1 THE ANOMALY THAT MOTIVATED THE STUDY



Figure 2: Left: pass@1 rises over 150 steps on all three runs, driven by first drafts (dashed) rather than final answers. **Centre: the count of items rescued by turn 2 or later never moves** — 6–9 out of 540 from step 0 to step 150, across two algorithms and two seeds. Right: the seeded-bug repair channel does improve, confirming the model is learning something about fixing code; it simply never does so *across turns*. §5.5 shows the tool was never successfully called in any of these runs.

G.2 RESULT TABLES REFERENCED FROM §5

Table 14: BFCL v4, first HTTP request per case. Intent columns are measured only where the parse failed.

Arm	cases	HTTP err	first parsed	unparsed	tight	strong	weak
hermes (documented)	200	0	0	200	166	169	187
qwen2.5-coder (repaired)	200	0	196	4	0	1	2

Table 15: τ -bench retail, all 115 tasks paired across arms (none dropped).

	documented	repaired
assistant turns	1389	1971
server-parsed calls	0	636
emitted but unparsed	789	6
tool executions / observations returned	0	636
tasks entering the tool loop at all	0	103
tasks solved	7	10

Table 16: Same probe, items and criterion under a *matched* envelope.

Size	server-parsed	silent fraction (tight, unparsed)
Qwen3-0.6B	86	1
Qwen3-1.7B	44	1
Qwen3-4B	70	2
Qwen3-8B	97	0
Qwen2.5-Coder 1.5B–32B (mismatched)	0 at every size	0 → 4 → 21 → 36 → 80

Table 17: Qwen2.5-Instruct under the identical mismatched configuration, with both ladders’ task outcomes measured on a working executor.

Size	Instruct, emitted			Coder		Instruct		
	parsed	tight	weak	parsed	turn-1 = final	turn-1	final	rescued
1.5B	1	0	7	0	31	13	13	0
3B	11	56	2	0	37	14	17	+3
7B	63	21	0	0	53	34	40	+6
14B	77	19	0	0	53	38	52	+14
32B	88	8	0	0	68	42	65	+23

Table 18: Llama-3.1-8B arms, paired on the same 100 items.

Arm	Calls issued	Wrong-tool	Executed	Turn-1	Final
terse (parser only)	97	23	74	34	49
+ official template	95	22	73	33	44
+ official template + strict	98	0	97	46	61
ReAct (reference)	98	0	97	61	80

Table 19: Decomposing the ReAct-FC gap on Llama, paired McNemar.

Comparison	b/c	<i>p</i>	Attributable to
terse-FC → strict-FC (49 → 61)	15/3	0.0075	interface repair, +12
strict-FC → ReAct (61 → 80)	27/8	0.0019	protocol, +19
terse-FC → ReAct (49 → 80)	38/7	3.1e-06	both, +31

Table 20: Repair loop on Qwen2.5-Coder-7B, `tool_choice: auto` fixed, changing only the adapter.

Arm	Parsed	Mean turns	≥ 2 turns	Turn-1	Final	Rescued
ReAct	95	1.92	40	57	74	17
FC + hermes	0	0.96	0	53	53	0
FC + dedicated adapter	84	1.82	37	53	62	9

Table 21: RL training rollouts, matched in model, data, hyper-parameters, seed and prompt strength.

Arm	Steps	num_turns/mean	Tool time	Tool calls / rollout
FC (<code>tool_agent</code> + hermes)	10	2.000 (constant)	0.00 s	0.000
ReAct (<code>react_agent</code>)	3	5.883	12.21 s	2.052

G.3 OFFLINE RE-PARSE MATRIX AND FAILURE-LAYER DECOMPOSITION

To exclude “your extractor is simply better than hermes”, we re-parsed the *same stored bytes* under four rules (`analysis/reparse_matrix.py`, no GPU).

Table 22: The same bytes, re-parsed offline under four rules. Where the server already parsed a call, `content` is empty and offline replay has nothing to read; those cells are N/A rather than zero. The gap is a parser mismatch, not a claim that no parser could have read the output.

Arm (server-parsed = 0)	server	hermes replay	<tools> replay	bare JSON	tight
Qwen-1.5B	0	0	0	0	0
Qwen-3B	0	0	0	5	4
Qwen-7B	0	0	0	22	21
Qwen-14B	0	0	0	42	36
Qwen-32B	0	0	0	100	80

The `hermes` column is zero at every scale: the model never produces that format. The bare-JSON and tight columns are not: the calls exist, and their difference from the server column is precisely what the format mismatch removes. For arms where the server *did* parse, offline content-only re-parsing is **undefined, not zero** — a successful parse moves the call into `tool_calls` and leaves `content` empty.

Replaying vLLM 0.27.1’s hermes extractor line for line over the stored bytes (analysis/failure_layer.py) separates the layer at which each item is lost: no `<tool_call>` envelope at all; envelope present but the JSON payload malformed; payload well-formed and rejected only because `json.loads` does not tolerate trailing text in the capture; or a genuine parser loss, meaning the replay succeeds where the server did not.

Table 23: Where the unparsed items are lost. Coder fails entirely at the envelope; Instruct reaches the envelope and fails at the payload. **Genuine parser loss is zero in every arm** — and across all 4254 de-duplicated first turns in the archive.

Arm	no envelope	payload malformed	strictness only	parser loss
Coder 1.5B–32B (all sizes)	100	0	0	0
Instruct 1.5B	93	6	0	0
Instruct 3B	22	41	26	0
Instruct 7B	3	33	1	0
Instruct 14B	13	6	4	0
Instruct 32B	1	9	2	0

The two ladders fail at different layers: Coder emits a well-formed bare call and never the envelope, so hermes not seeing it is *correct behaviour*; Instruct emits the envelope and then breaks the JSON, almost always by embedding Python whose docstring quotes or raw newlines are left unescaped inside the code string. And **in no arm — across the whole archive — did the server fail to parse a call that its own extractor would have accepted**. “Censoring” throughout this paper therefore means *the attempt never reaches the parser in a form it is defined to accept*, not *the parser discards valid calls*.

G.4 BFCL: GATING AND THE FULL REPRODUCTION RECORD

Each arm passes a two-case end-to-end smoke test before its full run, and a run is rejected unless the result count, the score file’s `total_count` and the number of distinct case ids in the recording proxy all agree at 200 with zero HTTP errors. Both arms run at `max_model_len = 16384`, `temperature = 0` and inference seed 0, on cases drawn at manifest seed 20260901. Registration of BFCL’s OpenAI handler happens in memory at import time, so `bfcl.eval`’s own files are left byte-identical; a recording proxy sits between benchmark and server and logs every request.

We ran the pair twice. The hermes arm is identical across runs in every column: 0/200 parsed, 166 tight, 480 requests, 0 errors. The repaired arm moves by one case between runs — **197 versus 196** first parsed, **99 versus 98** executed — while success is 19 in both. Greedy decoding does not make vLLM bitwise deterministic across differing batch compositions, and this arm runs eight threads through a multi-turn executor, so we report its intermediate funnel columns as ± 1 rather than exact.

Table 24: BFCL `multi_turn_base`, $n = 100$, joined to the HTTP trace by case id.

Arm	tight (unparsed)	first parsed	any parsed	executed	success	rescued
hermes	69	0	0	0	0	0
qwen2.5_coder	0	97	98	98	19	0

Table 25: Two quantified failure layers; the full four-family table is Table 26.

Family	Layer	Symptom	Detectable in advance?	Remedy
Qwen2.5-Coder	parser	emits a fenced JSON block, not <code><tool_call></code>	× template check false-positives	dedicated adapter
Llama-3.1-8B	schema	calls the <i>task function</i> as a tool (23%)	× requires argument inspection	<code>strict: true</code>

G.5 THE FULL FOUR-FAMILY TAXONOMY

Table 26: Four families, four layers. Only Mistral’s failure announces itself.

Family	Layer	Symptom	Detectable in advance?	Remedy
DeepSeek-Coder	template	chat template never injects tools	✓ template inspection	none
Qwen2.5-Coder	parser	emits a fenced JSON block, not <code><tool_call></code>	× template check false-positives	dedicated adapter
Llama-3.1-8B	schema	calls the <i>task function</i> as a tool (23%)	× requires argument inspection	<code>strict: true</code>
Mistral-7B-v0.3	token	repeats <code>[TOOL_CALLS]</code> → HTTP 400	~ errors, but only sometimes	none found

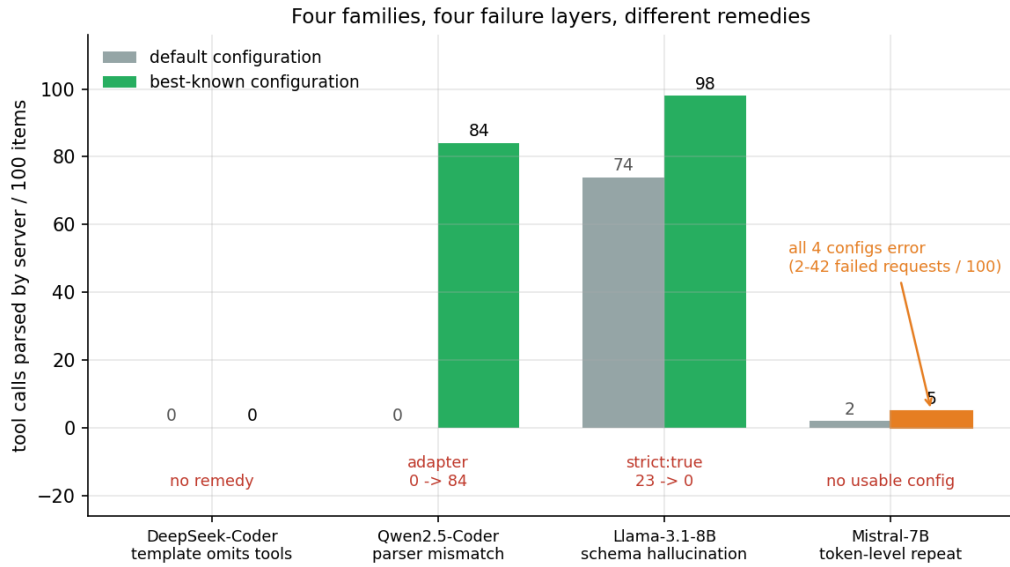


Figure 3: The same four failures, quantified. Grey is the configuration a reader arrives at by following the documentation; green is the best configuration we found. Mistral never produces an error-free run under any of four configurations, so its bars are not comparable with the others and are drawn in orange.

G.6 MAIN TABLE UNDER UNIFIED CONFIGURATION

Table 27: All arms genuinely at `max_tokens=2048`, one shared 100-item set, `temperature=0`. Mistral has no admissible FC arm: all four configurations produce request errors, so no pass rate and no *p*-value are computed.

Model	ReAct		FC		b/c	<i>p</i>	FC arm
	turn-1	final	turn-1	final			
Llama-3.1-8B	61	80	46	61	27/8	0.0019	official tmpl. + <code>strict</code>
Qwen2.5-Coder-7B	57	74	53	62	17/5	0.0169	dedicated adapter
Mistral-7B-v0.3	26	33	N/A	N/A	N/A	N/A	—

G.7 THE FOUR LLAMA CONTROLS, TABULATED

Table 28: Four single-variable controls on Llama-3.1-8B’s 23% wrong-tool rate. The first three rejected their alternative explanations and pointed at the model; the fourth found the actual cause and withdrew that conclusion.

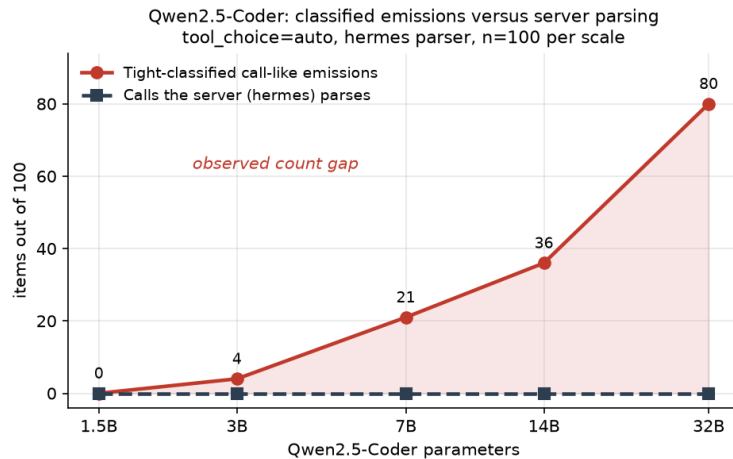


Figure 4: Within-family observation under the mismatched interface. Across the five Qwen2.5-Coder checkpoints, the server parses zero calls while tight-classified emissions rise to 80/100 at 32B (approximately 72 after calibration with tight-stratum precision 0.900, assuming transfer across sizes; Appendix F). Same data as Table 3.

Alternative explanation	Control	Empty-argument rate	Verdict
schema lacks a parameter description	rich vs terse	23 → 22 ($p=1.000$)	rejected
ReAct merely adds a reasoning step	FC + Thought scaffold	23 → 59 ($p<0.001$, worse)	rejected
server not configured per documentation	+ official chat template	23 → 22 (1/0 discordance, $p=1.000$)	rejected
schema constraint not enabled	+ strict: true on official template	22 → 0 (22/0, $p=4.77 \times 10^{-7}$)	accepted

G.8 CONDITIONING ON A SUCCESSFUL PARSE, AND ACCESS VS. USE OF FEEDBACK

Table 29: Conditioning on items where both protocols parsed successfully. The residual protocol gap holds on Llama and does not reach significance on Qwen — the honest reading is one family out of two, not a general law.

Family	Items both parsed	ReAct	FC	b/c	p
Qwen2.5-Coder-7B	83	63	56	11/4	0.119 (n.s.)
Llama-3.1-8B	96	78	60	25/7	0.0021

Table 30: Repair success given that feedback was actually received. Conditioned this way the two protocols are much closer, which locates the gap in *how often feedback arrives* rather than in what the model does with it. On Qwen this difference is within the non-significant band above, so we state it as an observation, not a result.

Arm	Reached turn ≥ 2	Rescued	Repair success given feedback
ReAct	40	17	42%
FC + adapter	37	9	24%
FC + hermes	0	0	undefined

Adapters are not a universal switch. The same adapter recovers 84/100 at 7B and **1/100 at 3B**: the 3B checkpoint ignores the `<tools>` few-shot entirely (zero `<tools>` tags in 100 items, 99 direct code) while still solving the task at a comparable rate (final 53). Adapter effectiveness is non-monotonic in scale, reinforcing the per-checkpoint claim of §5.3. The dedicated solution also

changes parser, chat template and few-shot simultaneously; we additionally rewrote the template’s few-shot examples, which originally invoked `get_weather` — a tool absent from our schema, which would have induced calls to a non-existent tool and inflated both initiation and empty-argument counts. Original and modified hashes are recorded.

G.9 PRESSURE ON A BROKEN INTERFACE MAKES THINGS WORSE

Two independent controls point the same way. Adding a reasoning scaffold to FC raised role confusion from 23 to **59** (§5.3). Replacing the optional tool instruction with a mandatory one on Qwen2.5-Coder-1.5B moved parser-accepted calls not at all — still 0/100 — while final pass fell from **31 to 15** and unparsable outputs rose from ~ 0 to 46/100. Neither intervention touches the interface, and both make the observable outcome worse. **When an agent appears reluctant to use tools, strengthening the instruction is the cheapest thing to try and the most likely to mislead** — it can degrade task performance while leaving the tool-call count at zero, which looks like a model that is both unwilling *and* incapable.

Table 31: Probe replication on the 1.5B model actually being trained, same mandatory prompt (n=100). Not one output names `run_tests`, so at this scale the missing calls are not a parser mismatch — there was nothing well-formed to censor.

Category	n	%
wrong name, <code>arguments</code> carry complete code	0	0%
wrong name, <code>arguments</code> are the task function’s parameters	36	69%
wrong name, executable code present elsewhere in the body	16	31%

Table 32: Increasing pressure on the 1.5B model. Stronger instruction and few-shot examples raise the number of recoverable code payloads but never produce a call that names the tool.

	recoverable calls	naming <code>run_tests</code>	final pass
mandatory prompt (baseline)	52	0	15
+ role-disambiguation few-shot	64	0	13

G.10 FORMAL 7B CHECKPOINT CURVE AND FINAL-LINEAGE AUDIT

All checkpoint evaluations use the same 542-item multi-turn manifest, have zero request failures, and attest the indicated native weight step. The accepted curve uses the original repaired steps 0–90 and resumed steps 120–150; the abandoned original steps 91–116 are excluded. The repeated step-90 evaluation after resumption gave 415/426/11 rather than 415/428/13 and is retained as test–retest evidence, not substituted into the formal curve.

Table 33: Complete formal 7B held-out curve. Each cell is turn-1 pass / final pass / rescued after turn 1, with $n = 542$.

step	broken FC	repaired FC
0	418 / 430 / 12	418 / 430 / 12
30	418 / 430 / 12	418 / 429 / 11
60	420 / 433 / 13	417 / 431 / 14
90	418 / 432 / 14	415 / 428 / 13
120	419 / 433 / 14	415 / 427 / 12
150	418 / 430 / 12	415 / 427 / 12

The policies did update. The accepted logs contain all 150 steps with non-zero actor gradient norms: 0.0120–0.0251 for broken and 0.00641–0.01599 for repaired; mean training rewards span 0.150–0.469 and 0.168–0.435, respectively. A direct checkpoint audit over all 392 LoRA tensors (80,740,352 parameters) finds 45,471,572 values changed from broken step 120 to 150 (56.3%, ℓ_2 delta 0.0787) and 45,296,056 from repaired step 90 to resumed step 150 (56.1%, ℓ_2 delta 0.1222). These comparisons establish updates to saved LoRA weights over the retained intervals; evaluation completion records separately attest the native weight step.

The event join also resolves the funnel’s apparent gaps. Broken FC has 19,203 generation events, 8,689 tight emissions and 10 accepted/executed/observed calls. Repaired FC has 33,367 generation events, 17,608 tight emissions, and 16,912 accepted calls in 16,844 accepted generation events. Forty generations contain multiple calls, producing exactly the 68-call surplus; all 16,844 dispatches join to executions and observations by request ID and assistant turn. Extraction records lack those identifiers; accepted-generation and dispatch counts match per trajectory, rather than through a per-generation join. Tight and accepted are different tests: 16,323 events satisfy both, 1,285 are tight but unaccepted, and 521 are accepted but not tight. The repository’s `analysis/p3.final.evidence.py` verifies all evaluation and event hashes, reconstructs the final lineage, and emits the per-file manifest; the server-side checkpoint audit is reproducible with `analysis/p3.checkpoint.lora.audit.py`.

G.11 HISTORICAL REACT POSITIVE CONTROL, TABULATED

RL training rollouts: under FC the tool is never successfully called
Qwen2.5-Coder-1.5B, verl GRPO, mandatory tool-use prompt on both arms

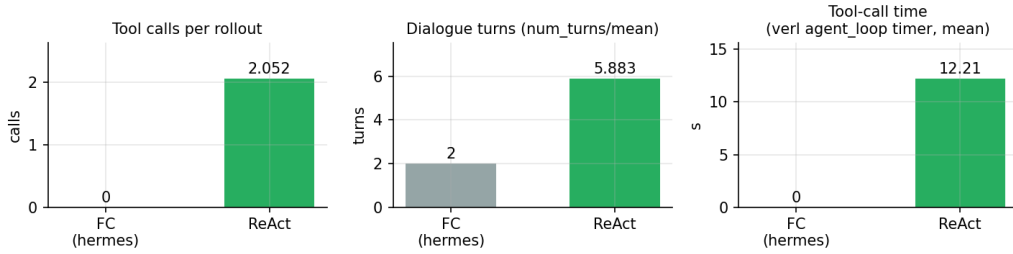


Figure 5: **Historical 1.5B RL diagnostic.** Under function calling the tool is never successfully called; `num_turns` stays pinned at its minimum of 2 and tool time is exactly zero. Prompt strength is matched across arms (both mandatory); step counts are not (10 vs 3). The formal 7B parser control is reported separately in Table 7.

Table 34: Historical ReAct training with a working tool channel, 23,676 executed calls. Rescues by $\text{turn} \geq 2$ do not move. The three historical 1.5B FC runs, in which **zero** tool calls executed, give rescue sequences [7, 9, 6, 6, 7, 6, 10, 8, 9, 8, 6], [8, 7, 6, 7, 7, 7, 5, 6, 6, 8] and [7, 5, 7, 5, 7, 8, 9, 9, 8, 9]. This is supporting evidence only; Table 7 is the single-variable control.

step	turn-1	final	rescued by $\text{turn} \geq 2$	repair channel
0	359	367	8	263
15	340	346	6	266
30	340	346	6	262
45	341	349	8	261
60	341	347	6	274

G.12 SAMPLING VARIANCE, AND AN INVERTED NOISE STRUCTURE

Llama-3.1-8B, `temperature=0.6`, `n=100 × 3` seeds.

Table 35: Sampling variance. FC seed 3 contains one rejected parallel-call request. We report its complete-denominator endpoint, and treat the two clean arms separately; an SD from two clean seeds would be too unstable to summarize variation reliably.

	seed 1	seed 2	seed 3	admissible
ReAct turn-1	53	56	43	3/3
ReAct final	72	72	72	3/3
FC turn-1	44	44	45	—
FC final	62	66	57 [†]	2 clean / 3 end-to-end

FC seed 3 contains one request error — the model emitted parallel tool calls and vLLM returned “This model only supports single tool-calls at once”. The end-to-end endpoint therefore retains that task as a failure and is 57/100; the prespecified clean-arm analysis excludes the seed and reports [62, 66]. An SD can be computed from two clean seeds, but would be an unstable estimate, so we do not use it to characterize sampling variation. Worst case among clean arms: ReAct’s lowest (72) versus FC’s highest (66) = +6.

Inverted noise structure. ReAct’s turn-1 varies widely (43–56, sd 6.8) while its final converges; FC’s turn-1 is nearly constant (44–45, sd 0.58) while its final disperses. The mechanism is compensation in the rescue count — 19 / 16 / **29**, largest where turn-1 was worst — and in all three seeds the count of “passed at turn 1 but failed finally” is **0**: the repair loop is monotone. **The three identical finals are a coincidence of totals, not identical item sets**, and sd = 0.00 must not be read as “ReAct is perfectly stable”.

G.13 MULTI-TURN REPAIR STABILISES THE TOTAL WHILE ITS COMPOSITION KEEPS MOVING

Two independent observations show the same structure. *Across sampling seeds:* ReAct’s final pass is 72 / 72 / 72 at temperature=0.6, yet the solved sets have pairwise Jaccard 0.71–0.80 — 15–24 items differ. *Across training steps:* between steps 15 and 30 of the ReAct run, 85 of 542 first-turn completions changed verbatim and 6 items flipped outcome (3 gained, 3 lost), leaving turn-1 and final pass numerically identical (340 / 346 at both checkpoints). The policy is demonstrably moving; the aggregate is not.

The common structure is that **the repair loop absorbs perturbation upstream of it**. Two consequences follow. *For measurement*, an aggregate pass rate is a **poor instrument for detecting change in a multi-turn agent**: it is stabilised by the very mechanism under study, and item-level set comparison detects movement the total conceals. We report both throughout, and note that had we reported only totals we would have concluded — twice, on two different axes — that nothing was happening. *For deployment*, stability of the total under sampling noise is a property practitioners want, it is contributed by the multi-turn loop rather than the base policy, and it is invisible in single-turn evaluation. **Bound:** both observations are on one model family and one task, the identical totals are in part coincidence, and we do not claim the aggregate is invariant — only that it is far less sensitive than its components.

G.14 THE TRAINING VERIFIER: SEAMS, AUDIT, AND PLANNED FIXES

The GRPO reward is outcome-only: it extracts the last submitted program, runs the item’s own pytest suite in the sandbox, and returns the pass ratio (`reward_code.py`). Three reward-hacking routes are closed by construction — a run in which no test executes scores 0 rather than 1, a timeout scores 0, and an item with no test string scores 0. The historical verifier nevertheless had two seams. (i) Its summary parser searched `(\d+)` passed over merged stdout/stderr and took the first match, so model stdout could imitate pytest’s summary. (ii) `skipped` and `xfail` were not counted in the denominator. Both were reachable in principle. We therefore audited: across the five training logs that retain generated code there are **zero** occurrences of a printed `“N passed”`, of `pytest.skip/allow.module_level`, or of `sys.exit/os._exit`, and `critic/rewards/mean` oscillates in 0.46–0.73 without the step change a discovered exploit produces. The audit’s own limit is that historical full rollout text was not retained, so it covers only the fragments those logs carry. Before the formal 7B control we anchored the regex to pytest’s summary, included `skipped/xfailed/xpassed` in the denominator, and retained every rollout in full with a SHA-256 digest. Both P3 arms use this fixed verifier. Neither verifier path is used for the reported held-out pass rates, which require `mode="script"`, a zero exit code and `status=="ok"`.

H LIMITATIONS, IN FULL

In the order a sceptical reader should apply them.

Scope of the evidence. (1) *One task family for the scale and training results.* Our own probe uses decontaminated KodCode items with a single tool taking one `code` argument. The external-validity

results are *not* code-only: BFCL’s `multiturn_base` spans file-system, vehicle and trading APIs, and `τ-bench_retail` is a customer-service domain whose 789 unparsed turns decompose to 100% envelope-layer with zero parser loss, reproducing the code result (§5.1). What remains code-only is the *scale ladder* and the *RL contamination* result. The former is a real gap — the same ladder on a non-code task would test whether the undercount’s growth with checkpoint size is a property of the interface or of code-shaped payloads (§6). The latter we do not expect to close: the reward is a `pytest` pass ratio, so porting it changes environment, reward and task at once. Multi-tool schemas and tool selection among alternatives remain untested throughout; a single-tool schema is an easy parser case, but we treat that only as motivation for broader tests, not as a quantitative lower-bound argument. (2) *Item sampling*. The 50 full-length arms use 100 items taken as `clean[:100]`, **not a random sample**; pairing across arms is exact, so within-study comparisons are sound. Generalisation rests on a replication on a random 300-item sample (seed 20260901): server-parsed remains 0 at every size across 1500 items, and strict emitted counts scale to 0, 4, 30, 40, 81 per 100 against the original 0, 4, 21, 36, 80. **7B moves from 21 to 30**, so the original subset understated that rung; direction and monotonicity are unaffected, but a single 100-item subset is evidently not a stable estimate of an intermediate rung. (3) *The scale result is within one family*. The $0 \rightarrow 4 \rightarrow 21 \rightarrow 36 \rightarrow 80$ ladder is Qwen2.5-Coder alone, against one mismatched interface, on one task; we do not claim censoring is generally scale-increasing, and we do not equate parameter count with capability. The Instruct ladder is a control on the *mechanism*, not a second ladder of undercount — its execution layer was initially void (the host lacked `pytest`) and has since been re-run with a working executor, the parse layer reproducing value for value. A second *family*’s ladder remains the test that would promote or refute the scale claim. (4) *The taxonomy is four instances, not an enumeration*. The evidence is deliberately asymmetric: Qwen has a scale ladder, a replay matrix and a repair; Llama a single-variable intervention; **Mistral has no admissible 100-item FC comparison at all**; DeepSeek is a template check with little quantification. (5) *Single-sample headline arms*. Table 20 and the ladders are one sample at `temperature=0`. Variance is estimated only on Llama-3.1-8B, and there only 2 of 3 FC arms are clean. A standard deviation from two seeds is defined but too unstable to summarize sampling variation reliably; the complete-denominator endpoint for the error-bearing seed is still reported (§G.12). (6) *Multiple comparisons*. Roughly a dozen McNemar tests; the Bonferroni threshold at that family size is ≈ 0.0042 . Surviving it: the official-template `strict: true` intervention on wrong-tool labels ($p = 4.77 \times 10^{-7}$) and final pass ($p = 7.63 \times 10^{-5}$), the protocol gap on 300 random items ($p = 3.1 \times 10^{-6}$), and the main effects of order 10^{-3} . **Not surviving it**: the repair-loop pass gain ($p = 0.093$ at $n = 100$, $p = 0.0385$ at $n = 300$), the Llama protocol contrast at $n = 300$ ($p = 0.0352$), and the residual-gap estimates ($p = 0.118, 0.125$). We rest no claim on the latter group, and report the $n = 300$ values precisely so a reader can see the direction is stable where significance is not claimed. (7) *Model scale stops at 32B*, on a single A800; no frontier models.

Threats to the causal reading. (8) *The clean repaired-FC control has one seed and one budget*. The completed 7B experiment holds model, LoRA configuration, data, prompt, decoding, verifier, seed and 150-step budget fixed, changing only the parser registered in `verl`. It therefore identifies the interface effect on the sampled experience distribution in this setting. Its null held-out result does not establish that a working channel is generally insufficient, and its LoRA training is not strictly comparable to the historical 1.5B full-parameter runs. (9) *Conditioning on a successful parse is post-treatment conditioning*. Whether an item parses is determined by the interface under test, so the both-parsed subset is not a random subsample and its contrast is not a clean causal estimate. On Qwen the residual is +8.4 pp, 95% CI $[-0.5, +17.4]$, $p = 0.118$: **we do not detect a residual difference, which is not the same as establishing there is none**. (10) *Near-zero observed frequency is not zero probability*. The broken 7B arm executes 10 calls rather than none; its instrumentation describes a sampled distribution, not the support of the policy. We therefore report a nearly closed channel (10 executions from 8,689 tight-classified emissions), not a mathematically unreachable action. (11) *Attempts and discards are separated only in the new run*. The formal 7B arms retain complete rollout text with hashes, parser verdicts, execution events and returned observations, so 8,689 emitted attempts and 10 accepted actions are directly distinguished. The historical 1.5B run did not retain full text. Its same-prompt replication finds no correctly named call, but that cannot retroactively identify what every historical rollout emitted. (12) *The null learning observation is setting-specific*. Parser repair restores 16,844 executed and observed calls over 150 steps without moving held-out rescues from 12. This is the requested single-variable control, but it is one seed, 7B, LoRA, one code task and one training horizon. The earlier ReAct arm (23,676 calls, flat rescues)

is directionally consistent but remains confounded by its protocol change. (13) *Human validation covers one criterion on one sample*. Two blind annotators on 98 stratified items give inter-annotator $\kappa = 0.967$ on the 62 items where both read identical bytes, and classifier-vs-gold-standard $\kappa = 0.871$ after third-party adjudication (Appendix F). Two residual limits: the sample is stratified by classifier verdict, so it estimates per-stratum precision and recall, not a population rate; and stored raw output is capped at 4000 characters, so a call emitted past that point is invisible to classifier and annotators alike, creating false negatives. Because the classifier also has false positives, the net bias of the raw emitted-call count is not identified. The *first* round’s apparent unreliability ($\kappa=0.713$) was an artefact of a 2400-character truncation in the annotation pack, not of the raters; that pack is superseded. (14) *The historical training verifier had two exploitable seams*. Its summary parser used the first unanchored `(\d+)` passed match and omitted `skipped/xfail` from the denominator. We found no exploit signature in the five historical logs that retain generated code, but those logs do not contain every full rollout. For the formal 7B control we anchored the regex to `pytest`’s summary, included `skipped`, `xfailed` and `xpassed` outcomes in the denominator, and retained every rollout in full with a hash. Both arms use the identical fixed verifier, so it cannot explain their contrast. Neither verifier version is used for the separately reported held-out pass rates (§G.14).

Environment dependence and provenance. (15) *Serving-stack version dependence*. Every measurement is against `vLLM 0.27.1` and `verl 0.9.0`, so parser behaviour, template defaults and the semantics of `strict: true` are properties of those versions. We pin versions, parser and template hashes, and model revisions for exactly this reason, and we have not replicated on a second serving stack (e.g. `SGLang`), so we cannot say which findings are `vLLM`-specific. (16) *Repository-revision dependence*. Chat templates ship inside `tokenizer_config.json` and can be changed by the publisher without a model change; the `Qwen2.5-Coder` false positive is precisely an artefact of an inherited template. (17) *Instrumentation errata, disclosed rather than absorbed*. Seven `ReAct` arms recorded `max_tokens=2048` while running at 1024 (an asymmetry that *favoured* FC; re-running changed final pass by ≤ 1 item); `Llama`’s initiation rate required manual correction from 97 to 74 after tool-name validation was added; five arms are error-bearing and admissible only for the error-rate census, and one previously reported $p=0.648$ computed on such an arm is withdrawn. All are enumerated in Appendix A. None is silently absorbed into a reported number.

WHAT WOULD CHANGE OUR CONCLUSIONS (RESTATED)

We name these so the claims are falsifiable rather than merely hedged. *Replications or a longer repaired-FC budget*. The requested broken-versus-repaired single-variable control is now complete: parser repair restores 16,844 tool interactions and does not increase held-out rescues. A consistent rescue gain over additional seeds or a longer preregistered horizon would narrow the scope of this 150-step, single-seed observation; continued nulls would strengthen it within, but not beyond, the tested model, task and optimiser setting. *A second family’s scale ladder*, with a working test executor: if the undercount does not grow with checkpoint size in another family, §5.2’s scale claim is a `Qwen-plus-hermes` fact rather than a scaling phenomenon. *The same ladder on a non-code task*. The five `Qwen2.5-Coder` checkpoints run against τ -bench `retail` under the documented configuration — models and interface fixed, only the domain varied. If the emitted column still rises with size while the parsed column stays at zero, the scale result is a property of the interface; if it does not, it is a property of code-shaped payloads. This needs a rescue arm at the small sizes for the reason `Granite` did: a checkpoint that simply never attempts a call produces an uninformative zero, not a censored one. *The same five-layer measurement on further standard benchmarks*. Done for `BFCL v4` and for τ -bench `retail` (§5.1), so the measurement-validity concern is *not* specific to our harness, and it now covers one interactive suite with a simulated user and stateful environment. What the τ -bench arm does *not* establish is an outcome effect: 7 versus 10 tasks solved on three discordant pairs is far from significant, and its user simulator is a local `Llama-3.1-8B` rather than `gpt-4o`, so its absolute scores are not comparable to the published leaderboard. Suites whose tool schemas are much larger, or whose scoring depends on argument-level matching, remain untested.